

Roadmap to Playwright with JS/TS + AI

AI Powered Playwright Automation Tester

Start here!

1. JavaScript /TypeScript

Javascript / Type Script

- Variables (let, const, var)
- Data types (Number, String, Boolean, Array, Object)
- Operators & Expressions
- Conditional Statements (if, else, switch)
- Loops (for, while, forEach)
- Practice exercises
- Functions & Arrow Functions
- Arrays and Objects (push, pop, map, filter, etc.)
- Hands-on coding practice
- Modules: Export and Import
- Template Literals
- Default Parameters & Destructuring

- What is TypeScript & why use it in automation
- Setting up TypeScript project with Playwright
- Types vs Interfaces
- Sync/Async Programming concepts
- Promises & Callback Hell Pyramid
- Understanding asynchronous code
- Optional Chaining (?.)
- Generics (basic)
- Promises and async/await pattern in TypeScript....

2. Playwright All concepts

Playwright (VSCode / Anti Gravity)

- Setting up VS Code for Playwright
- Essential VS Code plugins
- Workspace configuration
- Debugging in VS Code (Console output, Breakpoints, Watch variables)
- Finding & resolving errors
- Step into / Step over debugging techniques
- Configuring browsers (Chromium, Firefox, WebKit)
- Creating and running first test script
- Understanding configuration file
- Test fixtures, API Testing Context
- Playwright Inspector
- Trace Viewer
- Navigation (back, forward, reload)
- Multiple tabs/windows
- Practice browser operations
- Auto-waiting
- Persistent contexts
- Context configuration
- Cookies & local storage, Shadow DOM, SVG.

3. Advance Framework Concepts

Playwright Advance Framework

- Introduction to Test Frameworks
- Playwright Test Runner basics
- Test structure & organization
- Parallel execution
- Test groups & tags
- Retry mechanisms
- Assertions & Expect
- Test fixtures
- Before/After hooks
- Test reports
- Screenshots & videos
- Trace generation
- Configuration management
- Environment variables...

- Page Object Model (POM)
- Creating page classes
- Organizing test code
- Configuration layers
- Logging & reporting
- Error handling

4. AI Tools & Integration with Playwright

- Generative Framework via AI.
- Playwright MCP
- Playwright Agents
- 5+ Projects. (with End to end flow)
- Playwright + Stagehand
- Natural Language to Code.
- Ecom Project, API E2E Project
- Mixing Web + API Projects via utils.
- Docker with Sharding
- Utility functions
- Data-driven testing

- GitHub Actions
- Creating workflows
- Automated testing
- Jenkins integration
- Pipeline configuration
- Build automation



JavaScript/Typescript
Fundamentals



Playwright Fundamentals



PW Framework + CI/CD



Playwright + AI = Hero!